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RESEARCH ARTICLE

Using Machine Learning to Predict Wine Quality and Prices: A Demonstrative Case Using a Large Tabular Database

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ABSTRACT The wine industry is currently experiencing a worldwide growing interest, playing a substantial role in the economy of numerous countries. In this paper, we address the prediction of both wine quality and price, which is a valuable tool for several wine stakeholders, including producers, sellers and consumers. In particular, we explore a large worldwide tabular database that was retrieved from the *Vivino* platform and that includes around 167 thousand price records and more than 4 million quality examples. Using these data and a 10-fold cross-validation evaluation scheme, a large set of ML computational experiments was executed, considering three Feature Selection (FS) scenarios, up to six ML algorithms and five modeling approaches: a single (global) model, two divide-and-conquer ML modeling strategies (country and clustering based) and two hyperparameter tuning (for the single model and clustering filtering approaches). Overall, high quality predictive results were achieved. In particular, the price prediction best results were obtained by the clustering tuned FS1 scenario, while the best quality predictions were achieved by the clustering filtering FS3 setup.

INDEX TERMS Wine quality, wine price, machine learning, predictive models, dataset acquisition.

I. INTRODUCTION

The global wine industry has experienced significant growth in recent decades, though it faced setbacks due to the COVID-19 pandemic. *ReportLinker*¹ and *Statista*² reports vary in estimations on wine market value, with projections indicating a decline in value from 2019 to 2020 but a subsequent recovery, reaching a projected value of US\$434.6 billion by 2027. Despite revenue decreases in the short term, reports anticipate substantial growth in revenue and production volume between 2024 and 2029 [1].

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¹Available at <https://www.reportlinker.com/>

²Available at <https://www.statista.com/>

However, challenges persist, emphasizing the need for tools and insights tailored to the evolving landscape of the wine industry. While existing tools focus on production and market entry, there remains a gap in tools for evaluating wine quality and price, which is most likely due to the subjective nature of these assessments.

Wine quality is a subjective parameter influenced by various factors, including climate, soil conditions, wine-making processes, ageing, and bottling techniques [2]. This parameter is often assessed by wine critics, as well as through unlabeled wine tastings and community reviews [3], [4], aiming for an objective evaluation but inevitably involves inherent subjectivity [5]. Similarly, the price of wine is determined by both objective factors, such as production costs and profit margins, and subjective elements, such as wine

reviews and market positioning, that impact the perceived value [6]. Hedonic pricing models integrate these factors but may overlook experts' biases, with market positioning further influencing price, whether as an everyday or premium vintage [7], [8].

The subjective nature of traditional wine quality and price assessments motivate the need for a more objective approach. To address this, we propose using Machine Learning (ML) models to predict both wine quality and price, aiming to simplify the wine evaluation process, making it more accessible to a wider audience. In effect, the data-driven ML estimates can be used to support the valorization of wines, as performed by producers, sellers, experts or even consumers (e.g., is this wine good and fairly priced?). The ML predictions are particularly useful for unrated wine records or when new wines are introduced in the market, for which few expert knowledge or opinions are available. Finally, eXplainable Artificial Intelligence (XAI) methods can be used to extract relevant wine input attributes from the ML predictive models. Since some of these inputs can be controlled during the production process (e.g., alcohol levels), there is potential in using the extracted XAI knowledge to create better wines [9].

The main contributions of this research are:

- Unlike most of the related works, which are limited to specific regions and smaller datasets (as shown in Section II), our ML analysis encompasses a larger worldwide dataset that was recently collected from the popular *Vivino* platform and that includes around 167 thousand wine price records and more than 4 million of wine quality examples.
- We work with tabular data, assuming three sets of input feature scenarios: I – easily available at the wine bottle label (10 inputs; e.g., year, country); II – directly available the *Vivino* platform (total of 12 inputs; e.g., wine grapes); III – additional *Vivino* user inserted data (23 inputs; e.g., taste notes). The I predictive models can be directly used by producers or consumers, requiring only access to a wine bottle, while the II and III scenario based predictive models would require access to the *Vivino* service.
- We compare up to six ML regression algorithms to perform wine quality and price prediction, namely: Linear Regression (LR), Decision Tree (DT), Random Forest (RF), K-Nearest Neighbors (KNN), Gradient Boosting (GB) and Epsilon-Support Vector Regression (SVR).
- We compare several ML model approaches: single model – a unique ML is trained with all worldwide data; country filtering – assuming a divide-and-conquer approach, generating one ML model for each country (considering the top *Vivino* countries, namely the top 6 countries for price and top 10 countries for quality); clustering filtering – also assuming a divide-and-conquer approach, where a K-means clustering (via the elbow method) is used to generate K data clusters, each cluster being modelled by a ML algorithm.

This paper is organized as follows. Section II describes the related work. Then, Section III presents the collected data and

ML approaches. Next, Section IV reports the obtained results. Finally, Section V discusses the main conclusions and future work.

II. RELATED WORK

Table 1 summarizes the related works in terms of the following columns: **Study** - research work, chronologically ordered; **Features** - type of input data used to generate the predictive models; **Data Location (DL)** - addressed wine geographic region; **Data Dimension (DD)** - analyzed dataset size in terms of: wine examples (rows) $\times$ input variables (columns); predicted **Target Variable (TV)** – wine quality (Q), wine price (P) or both (B); and **ML Task (MLT)** – regression (R), classification (C) or both (B). To better contrast the related works with our research, the last row of Table 1 presents the main characteristics of this paper. We further note that in the table we do not include any predictive performance results, since these are specific to the type of data analyzed and the ML approach adopted.

When analyzing the **Features**, **DL** and **DD** columns it becomes clear that several studies analyze the same dataset. For instance, the known Wine Quality dataset was created in [9] and it includes 11 physicochemical tests (e.g., PH levels) related to two types of wine: white (4,899 samples) and red (1,599 instances). This dataset was explored in 18 of the research works shown in Table 1. Also, there are three studies that utilize the same Wine reviews dataset [13], consisting of 105,085 text reviews. In this work, we describe a new wine dataset that assumes recent tabular data retrieved from the popular *Vivino* platform. The dataset covers a large geographic region associated with 61 countries around the world. In Table 1, there are just two related works that used *Vivino* data [25], [35]. When compared with these two works, we have extracted a larger wine collection, with around 4 million distinct records. Turning to the predictive tasks (**TV**), most studies only approach wine quality prediction (26 works). There are four works that target wine prices. Finally, there are only two studies that perform both wine quality and price prediction (as our approach) [28]. Yet, these studies assume a textual analysis of wine reviews, which is rather distinct from our *Vivino* tabular data approach. Finally, a diverse set of ML algorithms is adopted by the related works. Most of the works (25 studies) only compare up to 3 distinct ML algorithms. In this paper, and similarly to [32], we explore up to 6 distinct ML algorithms. These algorithms are empirically compared under a large experimental study that includes different feature selection setups and different dataset filters (e.g., by country and by clustering).

III. MATERIALS AND METHODS

A. WINE DATA

Vivino is a popular wine-related platform. Unlike other platforms, *Vivino* offers open access to its data, ensuring unrestricted availability for researchers. Boasting a vast user base of over 68 million users, *Vivino* serves as a valuable platform that collects and disseminates wine-related knowledge, with an impressive database containing approximately 17 million wines. However, while the database encompasses a significant number of wines, only around

TABLE 1. Characterization of the related works.

Study	Features	DL	DD	TV	MLT	ML Models
[10]	Weather data	Bordeaux, France	36 x 7	Q	R	LR
[11]	Weather data	Bordeaux, France	10 x 8	P	R	LR
[9]	Physicochemical tests	Minho, Portugal	4,898 x 11	Q	R	SVR; NN; MR
[12]	Wine auction price records	France	100 x 4	P	R	ARMA
[13]	Wine reviews	World Wide	105,085 x 3	B	R	SVR
[14]	Barrel Data; Monthly mean Price	Bordeaux, France	-	P	R	Space State Methods
[15]	Experts ratings and weather data	Italy	136 x 8	Q	R	Ordered Probit Models
[16]	Physicochemical tests	Minho, Portugal	1,599 x 11	Q	R	XGBoost; LightGBM
[17]	Physicochemical tests	Minho, Portugal	4,899 x 11	Q	R	RF; KNN
[18]	Physicochemical tests	Minho, Portugal	2,312 x 11	Q	R	MR; LASSO; RF
[19]	Wine reviews	New Zealand	1,012 x 15	Q	R	NN
[20]	Physicochemical tests	Minho, Portugal	6,497 x 11	Q	B	RF; KNN; DT; NN
[21]	Wine bottle labels	World Wide	30,000 x 2	P	C	Naive Bayes; SVM
[22]	Wine ratings and reviews	World Wide	1,000 x 304	Q	C	DT; KNN; Naive Bayes; SVM
[23]	Physicochemical tests	Minho, Portugal	4,899 x 11	Q	C	NN
[24]	Wine reviews	World Wide	150,085 x 3	Q	C	SVM
[25]	Vivino Data	World Wide	124,397 x 1,000	Q	C	NN; DT
[26]	Physionomical tests	Minho, Portugal	4,898 x 11	Q	C	NN
[27]	Physicochemical tests	Minho, Portugal	4,899 x 11	Q	C	SVM; RF; KNN
[28]	Wine reviews	World Wide	105,085 x 3	B	C	SVM
[29]	Experts ratings and weather data	Germany	12,549 x 50	Q	C	Bagging CART; RF; C5.0; C4.5; PART; R Part
[30]	Physicochemical tests	Minho, Portugal	4,898 x 11	Q	C	LR; NN
[31]	Physicochemical tests	Minho, Portugal	4,898 x 11	Q	C	SVM; RF; NN
[32]	Physicochemical tests	Minho, Portugal	4,898 x 11	Q	C	Logistic Regression; SVM; DT; RF; Ada Boost Classifier; Gradient Boosting Classifier; KNN; Naive Bayes
[33]	Physicochemical tests	Minho, Portugal	4,898 x 11	Q	C	Naive Bayes; SVM; RF
[34]	Physicochemical tests	Minho, Portugal	1,599 x 11	Q	C	SVM; RF; DT; Genetic Algorithm
[35]	Vivino Data	World Wide	32,140 x 31	Q	C	RF; NN; Logistic Regression
[36]	Physicochemical tests	Minho, Portugal	4,898 x 11	Q	C	SVM; RF; KNN; NN, Logistic Regression
[37]	Physicochemical tests	Minho, Portugal	1,281 x 54	Q	C	XGBoost; RF; Extra Trees Classifier; Gradient Boosting Classifier
[38]	Physicochemical tests	Minho, Portugal	6,497 x 12	Q	C	NN; SVM; Logistic Regression
[39]	Physicochemical tests	Minho, Portugal	178 x 13	Q	C	SVM; NN
[40]	Physicochemical tests	Minho, Portugal	4,899 x 12	Q	C	LR; RF
This work	Vivino Data	World Wide	4,066,203 x 45	B	R	LR; RF; GB; KNN; SVR; DT

4 million of them are accompanied by user ratings, leaving numerous lesser-known brands without rating records [41]. Additionally, pricing data on *Vivino* relies primarily on web scraping tools, with each wine listing connected to a seller's website, resulting in approximately 2 million price records available, reflecting the inherent challenges in consolidating pricing information across various sources. The numbers consulted were accurate as of February 2024.

Given *Vivino*'s absence of an open API, we utilized a blend of web crawlers and API scraping techniques, mainly employing *Python*, *Selenium*, and the *Requests* package, to gather the necessary data. The data was collected during a six-month data period, from November 2022 to April 2023, resulting in 167,195 wine prices (ranging from 1 to 500 euros) and 4,066,203 wine quality (from 0.0 - lowest value to 5.0 – highest value) records.

B. DATA PREPARATION

The raw data comprised several categorical columns, which were converted into binary columns by using a one-hot encoding (when there are multi-labels, i.e., multiple nonexclusive labels that can be assigned to each wine) or a numeric label encoding (for single-label instances), as described in Table 3. Prior to advancing to the modelling stage, we applied a natural logarithmic transformation to the prices column. This transformation aimed to bring the variable values closer together, reducing the range and compressing the distribution. By applying the natural logarithm, we mitigated sparsity issues and achieved a more normalized distribution. This transformation also reduced the impact of extreme values and outliers, as illustrated in Figure 1.

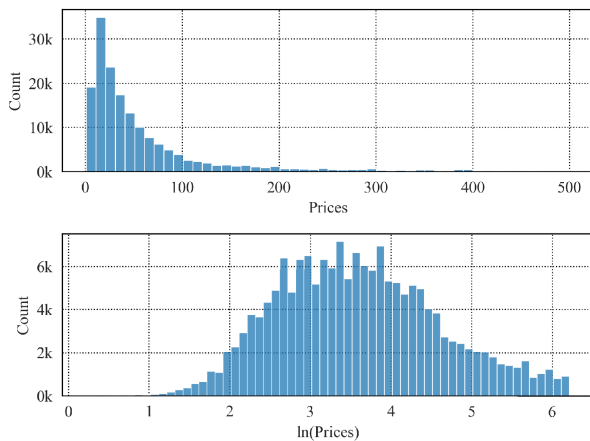


FIGURE 1. Histograms of the original wine price variable (top) and its natural logarithmic transform (bottom).

After transforming the data into a more usable format, the next step involved data cleaning and handling missing values, which were minimal in both datasets, reflecting the quality of the data from *Vivino*. The input missing values were addressed through imputation using the hot-deck method, namely an 5-NN imputer from *scikit-learn* module. Scaling was applied to ensure equal weight distribution during the imputation process. We use a Min-Max scaler with a range set between -1 and 1 . This scaler transforms the data, preserving the relationships between values while ensuring that they all fall within the specified range. The transform is obtained by computing the following equation:

$$x_{scaled} = (x - x_{min}) / (x_{max} - x_{min}) \quad (1)$$

where x_{max} and x_{min} denote the maximum and minimum attribute x value in the training data.

After preprocessing, both datasets retained their original number of records (167,195 for price and 4,066,203 for quality). Due to the one-hot encoding, the number of features increased to 174 inputs for the price dataset, while the quality dataset contains a total of 271 inputs.

C. EXPLORATORY DATA ANALYSIS

To gain insights into the wine data extracted, we conducted an initial Exploratory Data Analysis (EDA) focusing on two

key variables: Price and Quality. Fig. 1 and Fig. 2 display histograms for three variables: the original Price, its natural logarithmic transformed variant, and the original Quality. The top section of Fig. 1 illustrates the highly skewed distribution of original Price values, with most wines falling into lower price ranges. The transformation using a natural logarithm results in a more Gaussian distribution of the target variable (as shown at the bottom of Fig. 1). Regarding original Quality values, they exhibit a nearly Normal distribution with a slight negative skew, as depicted in Fig. 2.

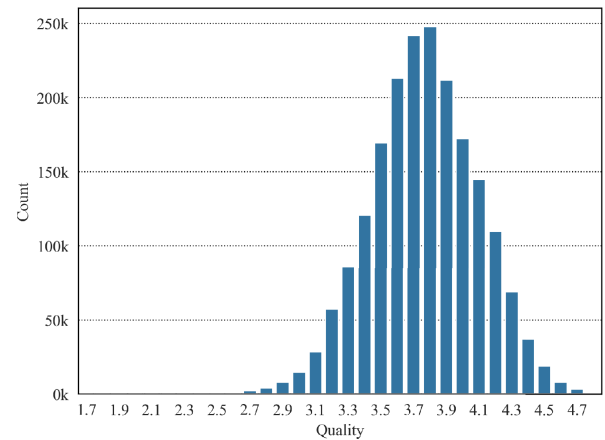


FIGURE 2. Histogram of the original wine quality variable.

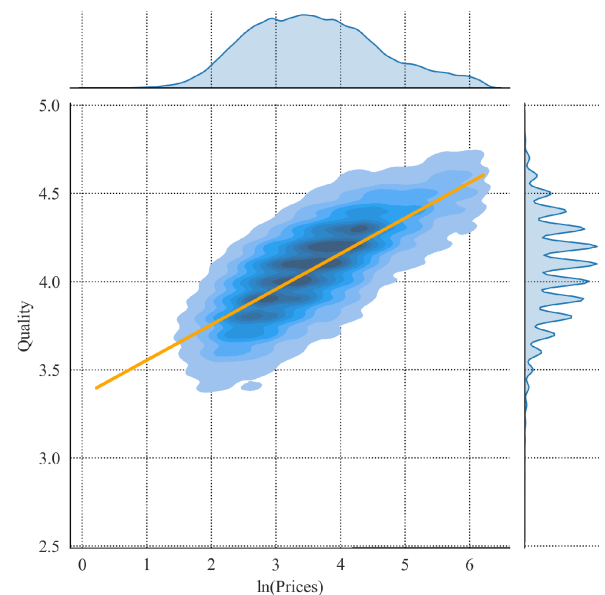


FIGURE 3. Density scatter plot of the target variables (x-axis - natural logarithm of the wine Price; y-axis - wine Quality ratings; the linear regression fit is shown by the orange coloured line).

We further conducted an EDA by exploring the relationship between collected inputs and the target variables. Illustrations of this EDA are provided in Fig. 3 and Fig. 4. As shown in Fig. 3, the relationship between the target variables is

nonlinear. The linear regression fit (orange line) indicates a positive effect between price and quality, as evidenced by a correlation of 0.74. However, this does not suggest a strong linear relationship, thus motivating the prediction of both variables independently. This approach allows us to capture the unique variations and trends in each relevant target, ensuring that any individual factors affecting them are appropriately accounted for. Furthermore, Fig. 4 shows box-plots depicting wine Quality across 61 analyzed countries. In the plot, the x-axis box-plots are arranged in ascending order based on the median wine Quality value for each country. The trend observed in Fig. 4 suggests that geographical location impacts the overall wine quality. Also, certain major wine-producing countries, such as Spain, Italy, and France, exhibit a considerable variation in the wine rating values. This observation prompted us to explore ML experiments filtered by the country attribute.

D. CONTEXT-BASED FEATURE SELECTION STRATEGY

As aforementioned in Section I, different stakeholders can benefit differently from our predictive tools:

- For **wine producers**, predictive models can aid in decision-making processes related to production planning and pricing strategies. By understanding the factors that influence wine prices and quality, producers can optimize their production processes, such as vineyard management, grape selection, and winemaking techniques, to enhance the overall quality of their wines. This can lead to improved market positioning and higher perceived value, ultimately increasing profitability. Additionally, predictive models can assist producers in setting appropriate pricing levels based on market demand, production costs, and quality indicators, enabling them to make informed pricing decisions to maximize revenue.
- **Wine sellers**, including wine retailers, distributors, and online platforms, can also benefit from predictive models. These models can provide insights into the expected market value of different wines, helping sellers optimize their pricing strategies and inventory management. By accurately predicting wine prices, sellers can ensure competitive pricing, avoid overstocking or understocking of wines, and optimize profit margins. Moreover, predictive models can assist sellers in identifying wines with high quality, enabling them to curate their product offerings to cater to consumer preferences and enhance customer satisfaction, something already very common in the food quality industry [42].
- For **wine consumers**, predictive models can serve as valuable tools for making informed purchasing decisions. By providing price predictions, consumers can assess the value for money when considering different wine options. They can also use quality prediction models to gain insights into the expected taste, aroma, and overall quality of a wine, helping them make choices that align with their preferences. Furthermore, predictive models can facilitate the exploration and discovery of new wines by suggesting alternatives based on

individual taste profiles, leading to enhanced consumer experiences and satisfaction.

Considering the diverse potential users, throughout the ML experimentation, we assume three distinct Feature Selection (FS) scenarios, each associated with a specific context of the data source:

- From the Wine product (e.g., label of a bottle). In practice, this context assumes wine characterization variables that can be easily accessed by inspecting the product.
- Directly available when using the *Vivino* platform. This context includes a more rich set of inputs.
- Similar to context II, except that the information needs to be manually inserted by wine consumers at the *Vivino* platform.

Table 2 illustrates the relationship of the FS scenarios with the data source context, while Table 3 enumerates the main features considered in each context.

TABLE 2. Adopted Feature Selections (FS) scenarios based on the data source context.

FS	Context		
	I	II	III
1	✓	✓	✓
2	✓	✓	
3	✓		

E. ML METHODOLOGY

Due to resource and time constraints, this paper focuses only on five main sets of ML experiments, illustrated in Fig. 5. These experiments include single model, country filtering, clustering filtering, and hyperparameter tuning (for single model and clustering filtering). Each experiment is conducted on both datasets, except hyperparameter tuning, which is applied only to the price dataset (since the quality dataset is too large, requiring a prohibitive computational cost for the hyperparameter tuning). The experiments address predictions for both price and quality, utilizing the previously described feature selection scenarios.

We utilize the popular *scikit-learn* Python library to implement the ML algorithms, including LR, DT, RF, KNN, GB and SVR. Data scaling was applied to the models that benefit from it (e.g., SVR) using the same Min-Max scaler referred to in Section III-B. We further note that the SVR model was only tested for the price prediction experiments, since the quality data is quite large, containing more than 4 million records. It should be noted that the SVR algorithm is computationally infeasible when trained with very large datasets (e.g., the computational memory and effort tend to scale quadratically with the number of training examples) [43]. In effect, due to memory and CPU power limitations of the adopted computational server, all SVR quality experiments halted and thus are not included in this work. The **single model** approach assumes the fit of one prediction model using all training data, using a

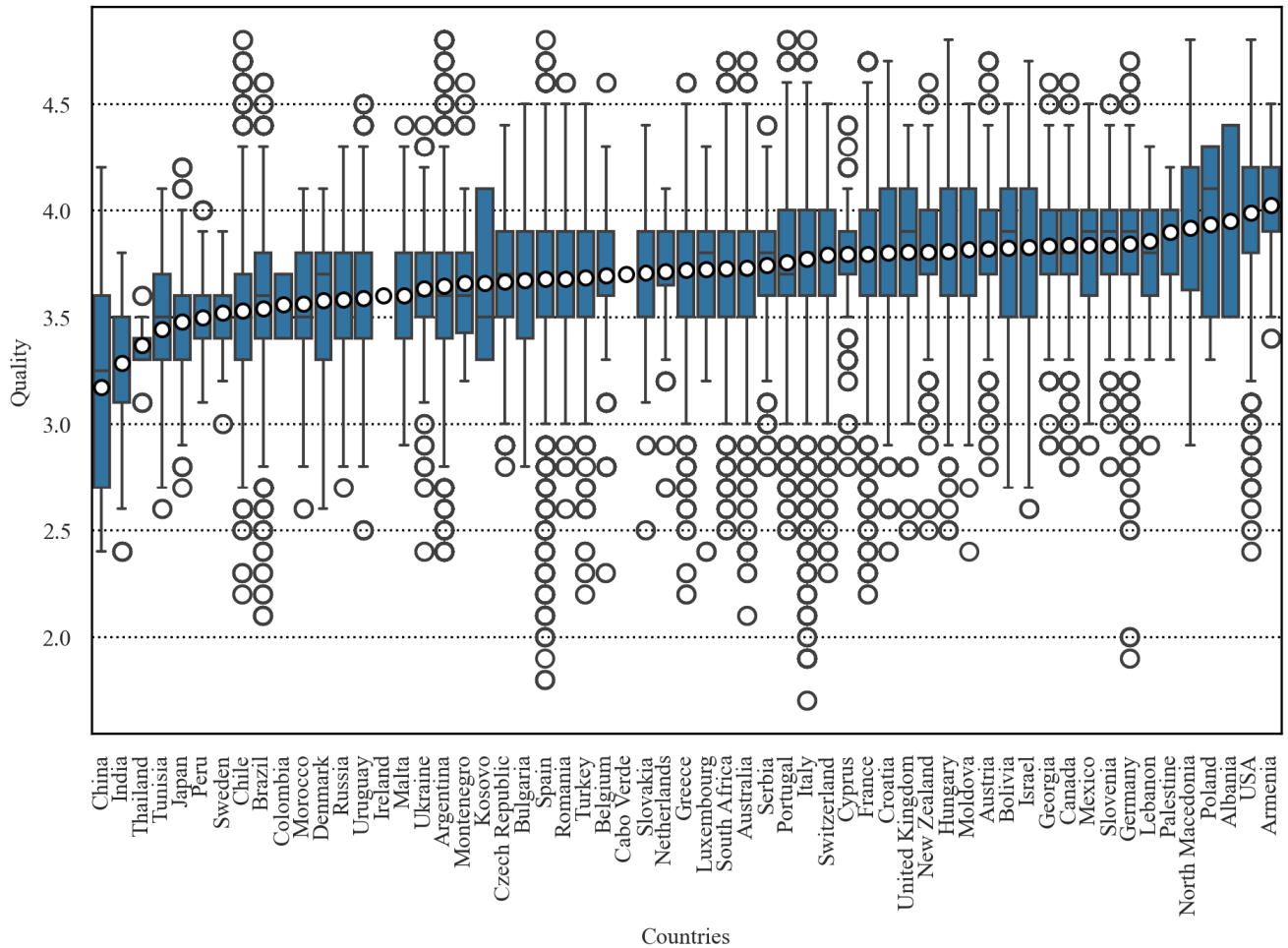


FIGURE 4. Box-plot of wine quality by country.

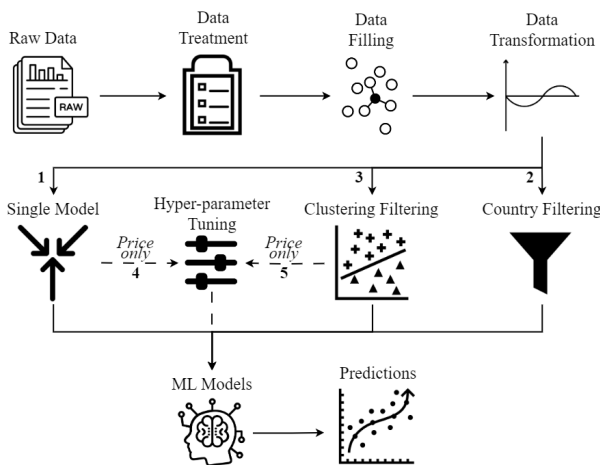


FIGURE 5. Adopted ML pipeline used in this research. Numbers represent the order of the experiments.

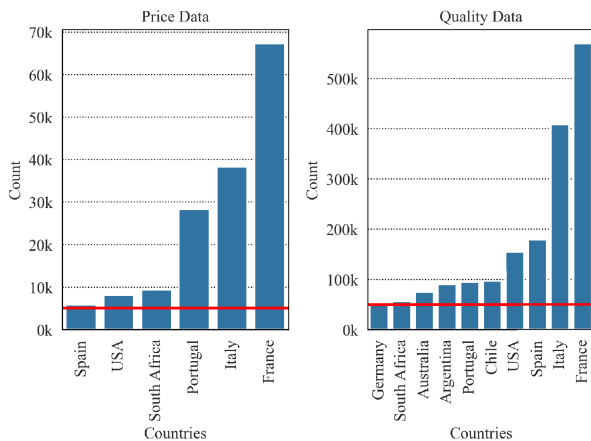
ML algorithm with its default hyperparameter values (as implemented in *scikit-learn*).

In the second set of experiments, **country filtering**, we explore a divide-and-conquer approach, in which a distinct ML model is fit to data filtered for a specific country. In order to have a minimum training set size, in this work we analyse countries with more than 5,000 Price or 10,000 Quality wine samples (the Quality dataset is substantially larger than the Price data), as shown in Fig. 6. Therefore, the country filtering involves the training of 6 distinct ML models for the wine price prediction (France, Italy, Portugal, South Africa, USA and Spain) and 10 specialized models for the quality target (same six countries and also Chile, Argentina, Australia, South Africa and Germany).

The objective of the third experiment was to explore an unbiased divide-and-conquer modelling approach, i.e., not linked to a specific input attribute but instead based on a **clustering** approach, namely via the K-Means algorithm, that was applied to all dataset inputs of a particular FS scenario. The K-means algorithm was directly applied to the price dataset while for the quality data (which is substantially larger), we chose a Mini-Batch K-Means clustering to effectively produce the clusters. In both prediction tasks, the

TABLE 3. Main features considered according to each context described.

Context	Source	Attribute	Description
I	General Wine Data	Year	Wine production year (number, range: [1749, 2022], e.g., 2012)
		Type	Wine type (categorical with 7 levels, encoding: label, e.g., 'red')
		Natural	Wine is natural (Boolean, e.g., True)
		Price	Wine Price (number, range: [1.25€, 500.00€], e.g., 14.99€)
		Style	Wine style (categorical with 624 levels, encoding: label, e.g., 'Riesling')
		Country	Wine Country (categorical with 60 levels, encoding: label, e.g., 'Portugal')
		Region	Wine Region (categorical with 2,725 levels, encoding: label, e.g., 'Douro')
		Winery	Wine Winery (categorical with 34,334 levels, encoding: label, e.g., 'Quinta do Vallado')
II	Vivino Data	Wine Dishes Pairing	Best dishes to accompany the wine (categorical with 22 levels, encoding: one-hot, e.g., 'Poultry')
		Wine Grape	Type of grapes used on the wine production (categorical with 109 levels, encoding: one-hot, e.g., 'Cabernet Sauvignon')
		Type Country	Origin country of wine type (categorical with 37 levels, encoding: label, e.g., 'Portugal')
		Type Region	Origin region of wine type (categorical with 15 levels, encoding: label, e.g., 'Douro')
III	Vivino User Data	Taste Notes	Taste notes sensed in wine (categorical with 13 levels, encoding: one-hot, e.g., 'oak')
		Ratings Average	Average of all ratings given to wine (Quality) (number, range: [1.7, 4.8], e.g., 4.5)
		Ratings Count	Total amount of ratings given to wine (number, range: [25, 103800], e.g., 377)
		Acidity	Sensed wine acidity by user (number, range: [1.0, 5.0], e.g., 3.4)
		Fizziness	Sensed wine fizziness by user (number, range: [1.0, 5.0], e.g., 1.2)
		Intensity	Sensed wine intensity by user (number, range: [1.0, 5.0], e.g., 2.3)
		Sweetness	Sensed wine sweetness by user (number, range: [1.0, 5.0], e.g., 2.7)
		Tannin	Sensed wine tannins by user (number, range: [1.0, 5.0], e.g., 3.0)
		User Structure Count	Total amount of structures given to wine (number, range: [0, 6930], e.g., 192)
		Wine Style Blurb	Sensed wine style blurb by user (categorical with 43 levels, encoding: label, e.g., 3)
		Wine Style Body	Sensed wine style body by user (number, range: [1.0, 5.0], e.g., 2.0)
		Wine Style Acidity	Sensed wine style acidity by user (number, range: [1.0, 3.0], e.g., 2.0)

**FIGURE 6.** Selected countries by considering a minimum of 5,000 Price samples (left) and 10,000 Quality samples (right).

optimal value for K was determined through an analysis of the resultant elbow curve. Depending on the FS scenario and ML algorithm, the best K value ranged from 2 to 6.

In the last set of computational experiments, we performed **hyperparameter tuning** for both single model and clustering filtering approaches with the price prediction target. The tuning was applied to the best performing models in the

single model (RF, GB and DT) and clustering filtering (RF, GB, DT and KNN) approaches. A grid search was executed, using a 10-fold cross-validation and the Mean Absolute Error (MAE) criterion to select the best hyperparameters. The search ranges are presented in Table 4.

TABLE 4. Hyperparameter ranges used during the grid search.

Hyperparameter	Range	DT	GB	RF	KNN
min_samples_split	2, 3, 5	✓	✓	✓	
max_depth	100, 500, null	✓	✓	✓	
min_samples_leaf	1, 3, 5	✓	✓	✓	
n_estimators	50, 100, 200, 500		✓	✓	
n_neighbors	3, 5, 7, 11, 17				✓
weights	uniform, distance				✓
algorithm	ball_tree, kd_tree, brute				✓

F. EVALUATION

In order to provide more information about the predictive performance, we compute 6 known regression predictive measures, namely: the Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), Mean Absolute Percentage Error (MAPE), Normalized Mean

Absolute Error (NMAE) and R-squared (R^2). Each of these metrics captures different aspects of the predictive performance, such as the magnitude of absolute (MAE and NMAE) or squared errors (MSE and RMSE), percentage errors (NMAE and MAPE) and the proportion of variance explained by the model R^2 . In this work, the main criterion for model selection purposes (e.g., when comparing the predictions of two ML models) is the best absolute error result (MAE or NMAE). If there is an absolute error tie, then the best squared error (MSE or RMSE) is adopted. The absolute error measures were selected since they are becoming popular in the forecasting domain [44]. Moreover, the best MAE results are often aligned with the best values obtained using other measures (e.g., R^2), as shown in Section IV. Finally, we note that all evaluation metrics were calculated on the original target scale values. For the ML price predictions, this involved the application of a post-processing that reverted the natural logarithm transform.

In order to obtain a robust generalization assessment of the ML predictive performances, a 10-fold cross-validation procedure was executed. Since the country and clustering filtering experiments produced several local ML models, the 10-fold procedure was executed for each individual model (e.g., using only French wine records for the France ML quality and price predictive models). In order to compare the single model and divide-and-conquer approaches, the obtained 10-fold test results were either filtered (single model) or aggregated (divide-and-conquer approaches). For instance, in Table 7 the local French wine predictions (country filtering approach) are directly compared with the 10-fold single model predictions filtered to the country of France. Similarly, in Table 8 the Cluster 0 predictive results are compared to the single model test predictions for the same Cluster 0 data records. As for the aggregated results, they are presented in Table 8 and correspond to a merge of the 10-fold test results from all divide-and-conquer ML models. In the case of the country filtering, the aggregated test set includes all data records from 6 (price) and 10 (quality) countries and it corresponds to the majority of the examples (thus termed “most samples”): 93.8% for the price dataset and 87.9% for the quality data. As for the clustering filtering, the aggregated test set includes all samples.

IV. RESULTS

The computational experiments were executed on a computational server using code³ written in the *Python* programming language and that adopted the *scikit-learn* and *pandas* modules. The obtained results are discussed in the next subsections.

A. SINGLE MODEL

The evaluation metrics for the complete 10 fold test predictions generated by each ML algorithm are presented in Table 5. In the table, the best predictive models are identified in terms of the best (lowest) MAE and NMAE measures

for each combination of Target Variable (TV, price – P or quality – Q) and Feature Selection (FS) scenario.

TABLE 5. Single model predictive results (best models are in bold).

TV	FS	ML Model	MSE	RMSE	MAE	MAPE	NMAE	R^2
P	1	DT	1244.07	35.27	13.74	0.33	2.29	0.78
P	1	RF	736.40	27.14	11.29	0.23	1.88	0.87
P	1	GB	1041.03	32.26	14.99	0.24	0.03	0.84
P	1	LR	2206.23	46.97	21.41	0.32	0.04	0.65
P	1	KNN	3045.20	55.18	28.00	0.87	4.67	0.47
P	1	SVR	6260.81	79.13	41.34	0.88	0.08	0.01
P	2	DT	3316.62	57.59	26.08	0.72	4.35	0.35
P	2	RF	1943.67	44.09	20.08	0.45	3.35	0.62
P	2	GB	3211.25	56.67	27.22	0.45	0.05	0.49
P	2	LR	4872.21	69.80	34.53	0.59	0.07	0.23
P	2	KNN	2541.71	50.42	24.46	0.67	4.08	0.50
P	2	SVR	6512.66	80.70	40.71	0.78	0.08	-0.03
P	3	DT	3566.57	59.72	27.12	0.75	4.53	0.30
P	3	RF	2122.90	46.07	21.05	0.47	3.51	0.58
P	3	GB	3400.99	58.32	27.98	0.46	0.06	0.46
P	3	LR	6790.77	82.41	40.85	0.80	0.08	-0.07
P	3	KNN	2581.96	50.81	24.80	0.69	4.14	0.49
P	3	SVR	6779.04	82.33	42.36	0.85	0.08	-0.07
Q	1	DT	0.01	0.10	0.02	0.01	0.01	0.92
Q	1	RF	0.01	0.08	0.03	0.01	0.01	0.95
Q	1	GB	0.06	0.25	0.20	0.05	0.06	0.41
Q	1	LR	0.10	0.31	0.24	0.07	0.08	0.11
Q	1	KNN	0.03	0.18	0.09	0.03	0.03	0.71
Q	2	DT	0.003	0.06	0.01	0.002	0.003	0.97
Q	2	RF	0.002	0.05	0.01	0.004	0.004	0.98
Q	2	GB	0.07	0.26	0.20	0.05	0.06	0.40
Q	2	LR	0.10	0.32	0.25	0.07	0.08	0.09
Q	2	KNN	0.03	0.17	0.09	0.02	0.03	0.75
Q	3	DT	0.0007	0.03	0.002	0.0004	0.0005	0.99
Q	3	RF	0.001	0.03	0.007	0.002	0.002	0.99
Q	3	GB	0.07	0.26	0.20	0.05	0.06	0.41
Q	3	LR	0.11	0.33	0.26	0.07	0.08	0.04
Q	3	KNN	0.007	0.09	0.02	0.006	0.007	0.93

1) PRICE

For the single model price prediction approach, the RF algorithm provided the best predictive results for all FS scenarios. Regarding the FS scenarios, the predictive performance always diminishes when fewer input features are available. Thus, the best results were provided by FS1, followed by FS2, and then FS3. More importantly, the best single model results for price prediction were of high quality, with the best predictive models producing a R^2 of 0.87.

2) QUALITY

For the single model approach, the DT model obtained the best quality predictions for all FS scenarios. In contrast to the price target, for the quality target, the feature scenario impact is the opposite, with the best results being obtained for the FS3 scenario (usage of inputs available at the bottle label).

³Several code examples of the predictive models are freely available at: <https://github.com/DMoreira97/ML-wine-quality-and-prices-prediction>

The best single model results for quality prediction were also of high quality, with the best predictive models producing a R^2 of 0.99.

3) COMPUTATIONAL EFFORT

The computational effort required by the single model ML algorithms is presented in Table 6. The table shows the average elapsed time (in seconds) required for each training iteration of the 10-fold procedure and the average time to predict the quality or price for a single wine (inference time, in microseconds).

Regarding the training effort, the GB and SVM algorithms require the highest computation times. Nevertheless, the required effort is still affordable given the large amount of training data involved. For instance, the most demanding price prediction model (GB) requires a training time that is less than 5 minutes, while for quality data (with millions of records), GB requires a maximum training time of around 76 minutes. More importantly, the training effort required by the best performing predictive models (see Table 5) is much more reasonable. For instance, the best RF price prediction model requires a training time that is less than 33 seconds, while the best DT quality prediction model demands only around 11 seconds of training time.

Turning to the inference time (predict a single wine instance), several of the tested ML algorithms are very fast. For instance, all DT, RF, LR, and GB fitted models require less than 15 microseconds to issue a prediction. Thus, these ML algorithms, in particular the selected single model RF (price) and DT (quality), can be used for real-time predictions.

B. COUNTRY FILTERING

For each country, we tested all six ML algorithms used in the single model approach. Consistent with the single model results (Section IV-A), the RF algorithm was the best for wine price prediction, while DT excelled in wine quality prediction. The performance of these models for the country-specific approach is shown in Table 7, which compares individual country test results (e.g., France, Italy) to the same country-wise predictions from the single model (Section IV-A).

The aggregated results for all countries are discussed in Section IV-D. To reduce the width of the table, Table 7 only displays three predictive measures: RMSE, MAE and R^2 .

1) PRICE

The country-specific results demonstrate that this ML divide-and-conquer approach generally yields better results. Specifically, in 13 out of 18 price prediction scenarios (72%), the country-specific approach produced the best results. Analysing Table 7 in detail, it is possible to observe that local evaluations significantly enhance the predictive ability in some countries. For instance, in FS2, USA wine price predictions improved by almost 5 EUR (MAE), and in FS3, Spain wines improved by nearly 2 EUR (MAE). Most of the other tested scenarios showed slight improvements in the price predictions. These better local results confirm

TABLE 6. Single model computational effort for each target variable (TV), feature selection (FS) and model (average training time for 1 fold execution and average time for a single prediction).

TV	FS	Model	Training	Prediction
			Time (s)	Time (μ s)
P	1	DT	2.41	0.50
P	1	RF	32.57	5.75
P	1	GB	263.95	16.50
P	1	LR	2.71	2.00
P	1	KNN	0.09	785.00
P	1	SVR	132.70	5970.50
P	2	DT	1.13	0.75
P	2	RF	17.67	5.25
P	2	GB	122.97	5.75
P	2	LR	1.04	1.50
P	2	KNN	0.07	631.50
P	2	SVR	110.86	5094.00
P	3	DT	0.37	0.25
P	3	RF	4.90	5.50
P	3	GB	47.82	3.50
P	3	LR	0.02	0.50
P	3	KNN	0.23	3.50
P	3	SVR	24.25	984.25
Q	1	DT	60.28	1.75
Q	1	RF	847.16	8.75
Q	1	GB	4566.60	11.00
Q	1	LR	15.20	1.00
Q	1	KNN	2.67	5347.00
Q	2	DT	42.15	1.25
Q	2	RF	624.17	12.75
Q	2	GB	3513.45	10.50
Q	2	LR	17.32	1.00
Q	2	KNN	2.44	4942.25
Q	3	DT	10.83	0.50
Q	3	RF	147.16	7.75
Q	3	GB	1096.49	3.75
Q	3	LR	0.36	0.25
Q	3	KNN	4.49	3.50

that country-based ML models tend to produce better price predictions when compared to models trained on worldwide wine data. Regarding the FS scenarios, the best price predictions were achieved using FS1.

2) QUALITY

For quality prediction, the country filtered models managed even greater improvements, getting 28 enhanced results out of 30 (93%), and these were more substantial compared to price prediction. For example, in FS1 and for French wines, there is a four-fold decrease in the predictive error, a three-fold for Italian wines, and this pattern repeats for most wines independent of the FS selected. As for the FS scenarios for quality predictions, all models performed similarly well, obtaining very low MAE values (around 0.01) and excellent

R^2 values (near 1.0), reaching improved results in all FS scenarios.

TABLE 7. Country filtering predictive results (best models per row are in bold).

TV	FS	Model	Country	Country Filtering			Single Model		
				RMSE	MAE	R^2	RMSE	MAE	R^2
P	1	RF	FRA	34.06	15.58	0.88	34.73	15.70	0.87
P	1	RF	ITA	24.03	9.56	0.87	24.57	9.77	0.87
P	1	RF	PRT	22.57	9.04	0.78	22.34	8.96	0.78
P	1	RF	ZAF	10.16	4.92	0.84	9.54	4.77	0.86
P	1	RF	USA	32.41	15.67	0.85	32.66	15.51	0.84
P	1	RF	ESP	22.53	9.32	0.87	21.96	9.05	0.88
P	2	RF	FRA	58.56	29.71	0.61	58.13	29.60	0.62
P	2	RF	ITA	42.84	19.00	0.53	44.64	19.93	0.49
P	2	RF	PRT	29.84	13.14	0.62	30.13	13.14	0.62
P	2	RF	ZAF	18.78	8.95	0.23	19.02	9.09	0.21
P	2	RF	USA	51.01	27.57	0.63	60.14	32.31	0.49
P	2	RF	ESP	39.24	17.44	0.55	40.84	18.38	0.51
P	3	RF	FRA	60.35	30.90	0.59	61.26	31.72	0.58
P	3	RF	ITA	43.37	19.20	0.52	45.38	20.30	0.47
P	3	RF	PRT	30.50	13.39	0.61	30.99	13.43	0.59
P	3	RF	ZAF	18.95	8.97	0.21	19.12	9.15	0.20
P	3	RF	USA	60.28	32.26	0.49	65.86	34.02	0.39
P	3	RF	ESP	39.69	17.69	0.54	43.16	19.45	0.46
Q	1	DT	FRA	0.04	0.005	0.98	0.09	0.02	0.92
Q	1	DT	ITA	0.07	0.01	0.95	0.12	0.03	0.87
Q	1	DT	ESP	0.06	0.008	0.96	0.07	0.009	0.96
Q	1	DT	USA	0.03	0.002	0.99	0.12	0.03	0.85
Q	1	DT	CHL	0.019	0.0009	0.99	0.03	0.003	0.99
Q	1	DT	PRT	0.02	0.001	0.99	0.11	0.02	0.89
Q	1	DT	ARG	0.03	0.002	0.99	0.04	0.003	0.99
Q	1	DT	AUS	0.07	0.01	0.95	0.08	0.01	0.94
Q	1	DT	ZAF	0.09	0.017	0.93	0.13	0.04	0.84
Q	1	DT	DEU	0.03	0.002	0.99	0.07	0.01	0.93
Q	2	DT	FRA	0.03	0.003	0.99	0.06	0.009	0.97
Q	2	DT	ITA	0.03	0.003	0.99	0.08	0.01	0.95
Q	2	DT	ESP	0.04	0.003	0.99	0.04	0.003	0.98
Q	2	DT	USA	0.02	0.001	0.99	0.08	0.01	0.93
Q	2	DT	CHL	0.01	0.0007	0.99	0.02	0.0009	0.99
Q	2	DT	PRT	0.01	0.0004	0.99	0.05	0.006	0.97
Q	2	DT	ARG	0.01	0.0005	0.99	0.02	0.001	0.99
Q	2	DT	AUS	0.05	0.006	0.97	0.06	0.009	0.96
Q	2	DT	ZAF	0.06	0.01	0.96	0.08	0.01	0.94
Q	2	DT	DEU	0.02	0.0009	0.99	0.05	0.006	0.97
Q	3	DT	FRA	0.01	0.0006	0.99	0.02	0.001	0.99
Q	3	DT	ITA	0.02	0.0007	0.99	0.03	0.002	0.99
Q	3	DT	ESP	0.02	0.0007	0.99	0.02	0.0006	0.99
Q	3	DT	USA	0.007	0.0001	0.99	0.03	0.003	0.99
Q	3	DT	CHL	0.005	0.0002	0.99	0.007	0.0002	0.99
Q	3	DT	PRT	0.005	0.0001	0.99	0.03	0.002	0.99
Q	3	DT	ARG	0.008	0.0002	0.99	0.006	0.0001	0.99
Q	3	DT	AUS	0.02	0.001	0.99	0.02	0.001	0.99
Q	3	DT	ZAF	0.03	0.002	0.99	0.04	0.003	0.99
Q	3	DT	DEU	0.009	0.0002	0.99	0.02	0.002	0.99

Country: FRA - France, ITA - Italy, PRT - Portugal, ZAF - South Africa, USA - United States of America, ESP - SPAIN, CHL - Chile, ARG - Argentina, AUS - Australia and DEU - Germany.

C. CLUSTERING

All ML algorithms were also tested for the clustering divide-and-conquer modelling approach. Similarly to the previously obtained results, RF was the selected model for the price predictions and DT the best algorithm for the quality predictions (best ML algorithms among the tested ones). Table 8 shows the respective predictive results for each local model (cluster), which are compared with the single model test prediction results (filtered for each cluster data). We note that the overall (aggregated) clustering approach results (for all 10-fold test samples) are presented and analyzed in Section IV-D.

TABLE 8. Clustering filtering predictive results (best models are in bold).

TV	FS	Model	Cluster	Clustering			Single Model		
				RMSE	MAE	R^2	RMSE	MAE	R^2
P	1	RF	1	27.32	10.80	0.81	26.99	10.71	0.81
P	1	RF	2	25.77	10.68	0.86	26.29	10.97	0.86
P	1	RF	3	28.42	13.47	0.92	29.83	13.69	0.91
P	2	RF	1	36.34	15.26	0.66	36.42	15.36	0.66
P	2	RF	2	43.82	20.38	0.58	45.24	21.11	0.56
P	2	RF	3	57.05	29.90	0.65	61.16	32.88	0.60
P	3	RF	1	49.38	24.04	0.63	50.72	24.64	0.60
P	3	RF	2	48.65	22.20	0.59	49.29	22.46	0.57
P	3	RF	3	43.24	22.03	0.64	44.26	22.67	0.62
P	3	RF	4	37.17	15.86	0.52	38.02	16.07	0.50
Q	1	DT	1	0.12	0.03	0.85	0.09	0.02	0.92
Q	1	DT	2	0.08	0.01	0.94	0.10	0.02	0.92
Q	2	DT	1	0.15	0.05	0.80	0.12	0.03	0.89
Q	2	DT	2	0.05	0.006	0.98	0.04	0.003	0.99
Q	2	DT	3	0.11	0.03	0.90	0.06	0.008	0.97
Q	2	DT	4	0.09	0.02	0.90	0.05	0.007	0.97
Q	2	DT	5	0.05	0.005	0.98	0.04	0.004	0.99
Q	2	DT	6	0.10	0.02	0.92	0.06	0.009	0.97
Q	3	DT	1	0.02	0.0008	0.99	0.02	0.0007	0.99
Q	3	DT	2	0.02	0.0007	0.99	0.02	0.0008	0.99
Q	3	DT	3	0.06	0.008	0.97	0.05	0.006	0.98

1) PRICE

In terms of the local analysis for Table 8 (for each cluster data), the results favour the divide-and-conquer ML clustering approach only for the price prediction target. In effect, the clustering approach obtained 8 best price prediction results (of a maximum of 10 FS and cluster combinations, thus 80%). In particular, there are scenarios with high performance enhancements. For example, for FS2 and cluster 3, the predictions improved by almost 3 EUR. However, most of the remaining improvements showed differences of only a few cents.

2) QUALITY

For the quality prediction, the divide-and-conquer ML clustering approach was less effective. Only 3 out of 11 quality prediction results showed enhanced performance when compared to the single model ML strategy, resulting

in a success rate of 27%. This suggests that the clustering method does not offer the same level of improvement for quality predictions as it does for price predictions.

D. OVERALL COMPARISON

Table 9 presents the best predictive models that were obtained for each distinct ML approach, which includes the hyperparameter tuning. The table shows the predictive measures (RMSE, MAE and R^2) for all 10-fold test examples.

TABLE 9. Comparison of the best predictive models for the distinct ML approaches (best results per predictive task and FS scenario are in bold).

TV	FS	Model	RMSE	MAE	R^2
Single Model					
P	1	RF	27.14	11.29	0.87
P	2	RF	44.09	20.08	0.62
P	3	RF	46.07	21.05	0.58
Country Filtering					
P	1	RF	27.29	11.43	0.88
P	2	RF	44.21	20.10	0.64
P	3	RF	45.58	20.76	0.62
Clustering Filtering					
P	1	RF	26.80	11.15	0.87
P	2	RF	42.73	19.35	0.64
P	3	RF	45.13	20.68	0.60
Single Model Tuned					
P	1	RF Tuned	27.04	11.24	0.87
P	2	RF Tuned	44.04	20.04	0.62
P	3	RF Tuned	46.05	20.99	0.58
Clustering Tuned					
P	1	GB, RF and KNN tuned	26.35	10.58	0.88
P	2	GB Tuned	39.59	17.87	0.69
P	3	GB Tuned	40.87	18.56	0.67
Single Model					
Q	1	DT	0.10	0.020	0.92
Q	2	DT	0.06	0.010	0.97
Q	3	DT	0.03	0.002	0.99
Country Filtering					
Q	1	DT (most samples)	0.05	0.007	0.98
Q	2	DT (most samples)	0.03	0.002	0.99
Q	3	DT (most samples)	0.02	0.001	0.99
Clustering Filtering					
Q	1	DT	0.09	0.020	0.94
Q	2	DT	0.05	0.010	0.98
Q	3	DT	0.02	0.001	0.99

1) PRICE

When considering the price prediction task, the best results were obtained for FS1. For this FS scenario, high quality and similar R^2 values (0.87 and 0.88) were obtained for the three compared non-tuned approaches: single model, country and clustering filtering. Turning to the analysis of the MAE and RMSE values, the best overall results were obtained by the

clustering approach (e.g., MAE of 11.15), followed by the single model (e.g., MAE of 11.29).

The aim of the hyperparameter tuning is to assess the degree of predictive performance improvement that can be achieved when optimizing the ML hyperparameters.

Based on the overall performance results, the tuning was performed on the two best overall price ML approaches, single model and clustering based, as shown in Table 9. For the single model approach, the best tuned ML algorithm was the RF, while for the clustering filtering it consisted in GB for FS2 and FS3 (for all clusters) and GB, RF and KNN for FS1 (one ML algorithm for each cluster). When compared with the default single model, the tuned RF resulted in improved MAE and RMSE values but kept the same R^2 levels for all FS scenarios. As for the clustering approach, and for all FS scenarios, the hyperparameter tuning obtained better predictive performances for all three measures (MAE, RMSE and R^2). In effect, the best overall price prediction models were obtained by this clustering tuned approach.

For demonstrative purposes, the left of Fig. 7 plots the observed versus the predicted values for the FS1 scenario and clustering tuned price prediction approach. While there is some dispersion in the plot, most points are closer to the perfect diagonal line, as confirmed by the respective Regression Error Characteristic (REC) curve [45] (right of Fig. 7). The REC curve has a quality Area Under the Curve value (AUC) of 88%. For example, the REC curve shows that more than 80% of the predicted wine prices are accurately predicted if a tolerance absolute error of 10 EUR is accepted.

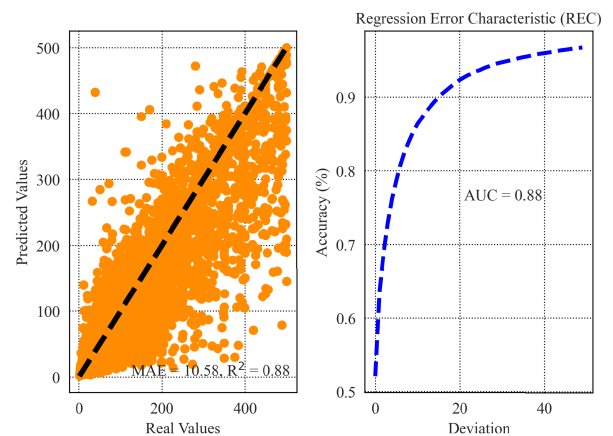


FIGURE 7. Clustering tuned FS1 results: left plot – measured (x-axis) versus predicted price values (y-axis); right plot – REC curve.

2) QUALITY

Regarding the quality task, all predictive models produced excellent predictions, with R^2 values near 1 and MAE and RMSE values close to 0. The best overall results were obtained for FS3 using both the country and clustering filtering approaches. While obtaining the same level of predictive performance, we suggest the usage of the clustering FS3 approach to obtain a global wine quality predictive model, since it requires less trained models (3 vs 10) when compared

with the country filtering approach. Moreover, it produces predictions for all tested wine samples, while the country filtering approach only covers 87.9% of the quality records (for the 10 top country models).

As previously explained, and due to computational costs, the hyperparameter tuning was only performed on the wine price dataset, which is much smaller than the wine quality data.

3) PRICE AND QUALITY

Having identified the best-performing models for each feature selection (FS), we proceed to compare their predictions. To facilitate this comparison, we have designed a Hexagonal Binned Plot, which visualizes Predicted Values versus Real Values for each model, as shown in Fig. 8. In the plots, it can be observed a pattern consistent with the error values previously obtained. For the price predictions (left of Fig. 8), the data points become less compact as the FS increases, indicating a wider spread in the predictions. Conversely, for quality predictions (right of Fig. 8), the data points become more compact with increasing FS, reflecting more accurate and consistent predictions.

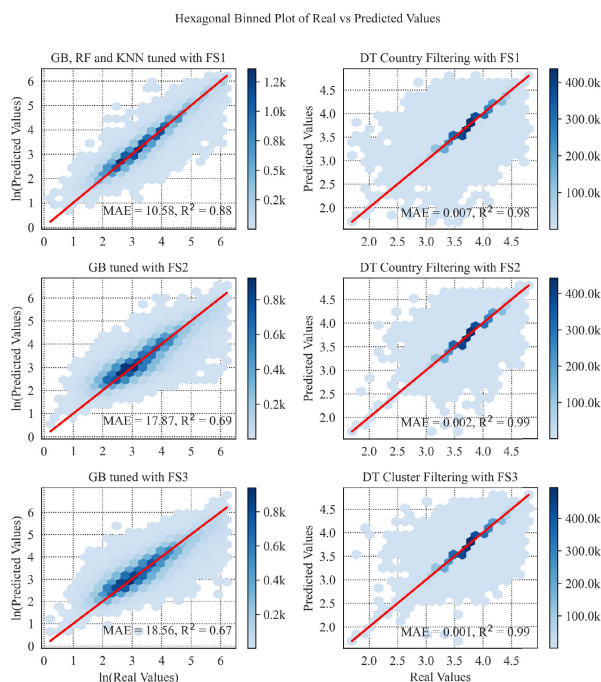


FIGURE 8. Predicted Values vs Real Values from each best-performing model with each feature selection plotted in a Hexagonal Binned Plot - natural logarithm of the Price values (x-axis) and predictions (y-axis) (right) and Quality values and predictions (left).

In general, the best predictive performances were obtained by the tree-based ML models, namely DT and two DT ensembles (RF and GB). One explanation for this behaviour is the categorical nature of most of the analyzed input features (see Table 3). In effect, tree-based algorithms tend to produce quality predictions when modelling categorical attributes [46]. And some of the adopted wine categorical features have a high cardinality (number of distinct levels),

such as the Wine Grape (109 levels), Region (2,725 labels) or Winery (34,334 levels). Another explanation is that the tree ensemble methods (RF and GB) can model more complex nonlinear phenomena. In particular, the RF algorithm tends to produce quality results with its default hyperparameter values, as shown by the empirical study performed in [47].

Concerning the input feature importances, we note that the predictive results displayed throughout this paper correspond to a large number of ML computational experiments. For instance, for the Single Model, the predictive results are shown as the merge of predictions obtained from a 10-fold cross-validation (10 distinct trainings). And for other modelling approaches, such as country-based filtering, the global results for a single cross-validation fold correspond to the union of several ML predictions (one for each country). Thus, due to lack of space, rather than providing full feature importance results, in this paper we present some demonstrative examples by adopting the SHAP XAI method [48]. We used the price dataset due to its more manageable size and took a model trained on all the data (Single Model) with all Feature Selections and a model trained just on data from France with Feature Selection 1 (FS 1), then compared their input feature importances as provided by their SHAP values, as displayed in Fig. 9.

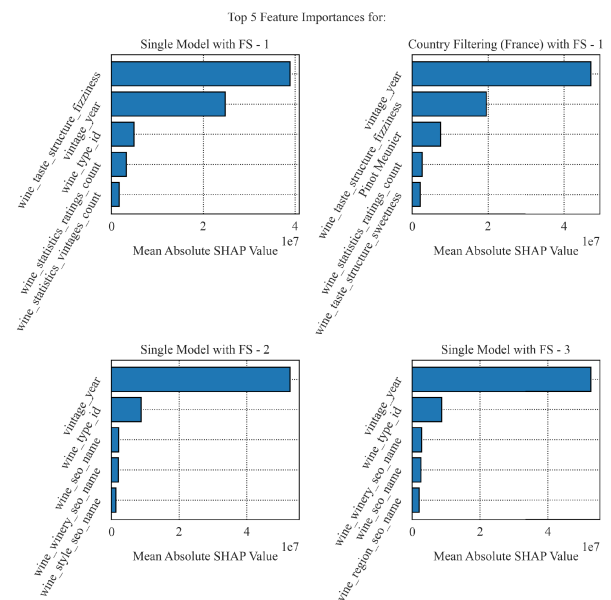


FIGURE 9. Feature importance calculated with SHAP when using the RF algorithm and some demonstrative ML modelling approaches.

As can be noticed, the vintage year is a very important feature across all scenarios depicted in Fig. 9. Wine type is an important feature too, showing a high importance in all models trained on the entire dataset, but not on the model trained just in data from France. In this model, the features generated from user reviews and a specific wine grape variety are revealed to be more impactful. “Pinot Meunier” is a red wine grape most noted for being one of the three main varieties used in the production of Champagne, thus with its origin in France. In the Single Model FS2, we can see that

besides data from the bottle label, the wine style also has an impact on the price of wine. Finally, in FS3, where only wine bottle label data is available, the winery and region show some importance.

V. CONCLUSION

The prediction of wine quality and price is a vital element to support the decisions of different stakeholders, including producers, sellers and consumers. Within this context, this paper proposes a Machine Learning (ML) approach to predict both wine quality and price, assuming a large worldwide tabular database (retrieved from the popular *Vivino* platform) that includes around 167 thousand price samples and more than 4 million quality records. Several ML experiments were held, considering a 10-fold cross-validation, three main Feature Selection (FS) scenarios, up to six ML algorithms and five modelling approaches (single model, country and clustering filtering, tuned single model and tuned clustering filtering).

In general, the divide-and-conquer approaches (country and clustering filtering), provided better local learning results (e.g., for a particular country such as France) when compared with the single model approach. The exception was the quality prediction task, where the single model approach outperformed the clustering filtering for 8 of 11 analyzed local clusters. Turning to the global aggregated results (considering all 10-fold test samples), high quality predictions were obtained. The best price prediction results were obtained by the clustering tuned approach using FS1 (all inputs), corresponding to a Mean Absolute Error (MAE) of around 10.58 EUR and a R^2 of 0.88. As for the quality prediction, the best results (considering all quality test samples) were provided by the clustering filtering FS3 approach. In effect, this approach achieved an excellent predictive performance (MAE of 0.001 and R^2 of 0.99).

Drawing conclusions from our strategy described in Section III-D, the results of our study hold significant practical implications for various stakeholders within the wine industry. Wine producers (FS1) can leverage the insights from our predictive models to enhance their decision-making processes, optimizing production and pricing strategies to increase profitability and market competitiveness. Wine sellers (FS2), including retailers and distributors, can utilize these models to refine their pricing strategies and inventory management, ensuring they meet market demand while maintaining optimal stock levels. Additionally, by identifying high-quality wines, sellers can curate their offerings to better align with consumer preferences, ultimately boosting customer satisfaction. For wine consumers (FS3), the models provide a valuable tool for making informed purchasing decisions, assessing the value for money, and exploring new wine options based on individual taste profiles. Overall, the context-based feature selection strategy we employed allows each stakeholder to utilize the most relevant data for their specific needs, thereby maximizing the benefits derived from our predictive tools.

In future work, we wish to explore deep learning algorithms to model our tabular *Vivino* wine database, such as

the deep multilayer perceptron architecture. It would also be interesting to explore Multi-Target Regression (MTR) approaches in which a single ML model is trained to predict both wine price and quality outputs. Moreover, we plan to have access to a larger computational infrastructure (e.g., with a large RAM memory and several CPUs and GPUs). Using such computational environment, it would be interesting to adopt a distributed computing and big data technology. This would substantially reduce the computational effort and increase the number of ML computational experiments (e.g., hyperparameter tuning for the wine quality predictive models). We also wish to enlarge the ML experiments by considering the extraction of more input features, such as the number of *Vivino* user ratings for a particular wine, wine bottle label images and text. Using such richer set of inputs, we intend to explore other advanced deep learning ML algorithms (e.g., convolutional neural networks, text transformers and Large Language Models (LLMs)).

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